

Sze Chai Leung (Mickey)

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EDUCATION

California Institute of Technology (Caltech)

September 2022 - present

Ph.D. Candidate in Mechanical Engineering

M.S. in Mechanical Engineering (June 2024)

University of Illinois at Urbana-Champaign (UIUC)

August 2018 - May 2022

B.S. in Mechanical Engineering with a Minor in Computer Science

- **GPA:** 3.98/4.00

PUBLICATIONS & PREPRINTS

1. **Leung, S. C.**, Zhou, D., & Bae, H. J. (2026). Reinforcement learning control of gust-induced airfoil lift fluctuations. *Manuscript in preparation*.
2. **Leung, S. C.**, & Mowlavi, S. (2026). FoamGym: an RL control library for the OpenFOAM PDE solver. *Manuscript in preparation*.
3. **Leung, S. C.**, Zhou, D., & Bae, H. J. (2026). Data-driven sensor placement for predictive applications: a Correlation-Assisted Attribution Framework (CAAF). *Communications AI & Computing*. 1, 8.
<https://doi.org/10.1038/s44488-026-00011-1>
4. Liu-Schiaffini, M., Singer, C. E., Kovachki, N., **Leung, S. C.**, Bae, H. J., Azizzadenesheli, K., & Anandkumar, A. (2026). Tipping point forecasting in non-stationary dynamics on function spaces. *Accepted at Proceedings of the National Academy of Sciences (PNAS)*.
<https://arxiv.org/abs/2308.08794>
5. Huang, X., **Leung, S. C.**, & Bae, H. J. (2026). Consistency requirement of data-driven subgrid-scale modeling in large-eddy simulation. *Physical Review Fluids*, 11(1), 014602.
<https://doi.org/10.1103/yykb-6rvf>
6. **Leung, S. C.**, Zhou, D., & Bae, H. J. (2024). Integrated gradients for optimal surface pressure sensor placement for lift prediction of an airfoil subject to gust. *AIAA Aviation Forum and ASCEND 2024* (p. 4148). <https://doi.org/10.2514/6.2024-4148>
7. Huang, X., **Leung, S. C.**, Whitmore, M. P., Elnahhas, A., & Bae, H. J. (2024). Consistent data-driven subgrid-scale model development for large-eddy simulation. *Proceedings of the Summer Program* (pp. 395–404). Center for Turbulence Research, Stanford University.
https://web.stanford.edu/group/ctr/ctrsp24/v04_HUANG.pdf
8. Ha, J., Kim, Y. S., Li, C., Hwang, J., **Leung, S. C.**, Siu, R., & Tawfick, S. (2023). Polymorphic display and texture integrated systems controlled by capillarity. *Science Advances*, 9(26), eadh1321.
<https://doi.org/10.1126/sciadv.adh1321>

WORK EXPERIENCE

Mitsubishi Electric Research Laboratories

Cambridge, MA

Research Intern

June 2026 – September 2026

- Established an RL control framework interfaced with the “OpenFOAM” PDE solver
- Demonstrated the framework across multiphysics flow control tasks

Department of Mechanical and Civil Engineering, Caltech

Pasadena, CA

Graduate Teaching Assistant, Fluid Mechanics/Turbulence

September 2023 – December 2023

- Delivered review lectures, conducted office hours, and managed assignments and exams

Mindray Medical International Limited

Mechanical Development Engineering Intern

Shenzhen, China

June 2020 - August 2020

- Led the design project of a single-person loading device for more convenient transportation of ultrasound machines
- Conducted mechanical testing on ultrasound prototypes and constructed engineering drawings

RESEARCH EXPERIENCE

Bae Research Group (<https://bae.caltech.edu/>)

Pasadena, CA

Advisor: H. Jane Bae, Assistant Professor of Aerospace

September 2022 – present

Data-Driven Flow Control & Sensing for Gust Mitigation on Airfoils

- Apply deep reinforcement learning (DRL) and conduct large-eddy simulations (LES) to design active control strategies that mitigate airfoil lift fluctuations
- Proposed a deep learning-based attribution framework to optimize sensor placement for target predictions across complex, nonlinear systems

Data-Driven Turbulence Modeling

- Developed subgrid-scale stress models via sparse regression coupled with a neural network to correct residuals and numerical errors, enhancing model consistency in LES

Ewoldt Research Group (<https://ewoldt.mechanical.illinois.edu/>)

Urbana, IL

Advisor: Randy H. Ewoldt, Alexander Rankin Professor

May 2021 – July 2022

Neural Network Constitutive Models for Non-Newtonian Fluids

- Formulated and fine-tuned neural networks to learn the physical properties of complex fluids from simulated flow fields

Kinetic Materials Research Group (<https://tawfick.mechse.illinois.edu/>)

Urbana, IL

Advisor: Sameh Tawfick, Professor; Ralph A. Andersen Faculty Scholar

May 2021 – October 2021

Capillary-Controlled Micro-Fin Morphing

- Tested and modeled the morphing behavior of polymer micro-fins under varying conditions

CONFERENCE PRESENTATIONS

1. **Leung, S. C., Zhou, D., & Bae, H. J. (2026).** Optimizing sensor placement in turbulent flows: a correlation-aware attribution framework. *3rd ERCOFTAC "Machine Learning for Fluid Dynamics" workshop*, Amsterdam, Netherlands.
2. **Leung, S. C., Zhou, D., & Bae, H. J. (2025).** Optimizing sensor placement in turbulent flows: a correlation-aware attribution framework. *The 78th Annual Meeting of the APS Division of Fluid Dynamics*, Houston, TX.
3. **Leung, S. C., Zhou, D., & Bae, H. J. (2024).** Integrated gradients for optimal surface pressure sensor placement for lift prediction of an airfoil subject to gust. *AIAA Aviation Forum and ASCEND 2024*, Las Vegas, NV.
4. **Leung, S. C., Huang, X., & Bae, H. J. (2024).** Consistent subgrid-scale model development for large-eddy Simulation with data-driven sparse identification. *The 77th Annual Meeting of the APS Division of Fluid Dynamics*, Salt Lake City, UT.

SKILLS

- Python, Fortran, MATLAB, C/C++ programming
- High-fidelity simulations (LES), OpenFOAM
- RL, deep learning, explainable AI
- Turbulence modeling, parallel computing on HPCs, mesh generation
- ProE/Creo, Solidworks and Fusion 360 CAD modeling